

CHAPTER 10

Sound Waves: Characteristics and Applications

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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🤔 Think It Over

1 **Two astronauts are repairing the arm of a space station together during a spacewalk. Can they talk to each other and hear the sounds of metal clanking as they do on the Earth?**

✔ ANSWER

No. Outer space is a near vacuum, and sound needs a medium (solid, liquid or gas) to travel.

So they cannot directly hear each other or the clanking of metal. They talk through special radio devices fitted in their spacesuits.

2 **How do most bats use sound to locate their prey in the dark at night?**

✔ ANSWER

Bats send out short bursts of **ultrasonic waves**. These waves reflect from nearby objects and prey.

By sensing the echoes, the bat finds the position of obstacles and prey. This is called **echolocation**.

⭐ 10.1 Production of Sound · Activities

Activity 10.1 **Stretch a rubber band across an open cardboard box and pluck it. Do you hear a sound? Is it vibrating? Do you**

hear sound after it stops? What happens when you change the tension? Is sound produced when plucked between two fingers? Is it as loud?

✔ ANSWER

- Yes, a sound is heard and the rubber band is seen vibrating.
- When it stops vibrating, the sound also stops.
- Stretching it more (more tension) gives a higher, shriller sound; loosening it gives a lower sound.
- Between two fingers, a sound is still produced but it is much softer than with the box.

Conclusion: Sound is produced by vibrations.

Activity 10.2 **Strike a tuning fork on a rubber pad and bring it near your ear. Do you hear a sound? Touch the water surface with a vibrating prong: do you see waves? Do you hear the sound in different orientations?**

✔ ANSWER

- Yes, a sound is heard.
- Waves form on the water surface, so the prongs are vibrating.
- Yes, the sound is heard in all orientations, so sound spreads in many directions.

Conclusion: Sound is produced by vibrating objects.

⭐ Pause and Ponder · 10.1 Production of Sound

1 **Explore various ways of producing sound.**

💡 SAMPLE ANSWER

- Plucking a stretched string (sitar, guitar, rubber band)
- Striking an object (drum, bell, taal, metal plate)
- Blowing air into a pipe (bansuri, whistle)
- Rubbing or scraping (bow on a sarangi string)
- Vibrating vocal cords (speaking, singing)

In every case, something vibrates.

2 **Make a list of different types of musical instruments and identify their vibrating parts which produce sound.**

💡 SAMPLE ANSWER

Instrument	Vibrating part
Sitar, veena, tanpura, ektara	Stretched strings
Tabla, dholak, mridangam	Stretched membrane
Bansuri, shehnai, harmonium	Air column (and reed in shehnai/harmonium)

Taal, ghanta (bell), jaltarang	Metal or the bowl and water

⭐ 10.2 Propagation of Sound · Activities

Activity 10.3 **Your friend knocks or scratches gently on a desk. Listen through air, then with your ear on the desk. Are you able to hear the sound through the table?**

✔ ANSWER

Yes. The sound is heard more clearly (louder) with the ear on the desk.

Conclusion: Sound can travel through solids.

Activity 10.4 **Tap two metal spoons together in air, then under water in a bucket. Do you again hear the sound? If sound did not travel through liquids, would you have heard this sound?**

✔ ANSWER

Yes, the sound of the submerged spoons is heard.

If sound could not travel through liquids, we would **not** have heard it.

Conclusion: Sound can travel through liquids (water) as well.

? **Would you hear sound in vacuum?**

✔ ANSWER

No. In the vacuum bell jar experiment, the sound becomes fainter as air is pumped out and is almost not heard in near vacuum. Sound needs a medium to travel.

⭐ Pause and Ponder · 10.2 Propagation of Sound

3 **Assertion (A): We cannot hear the sound of a bell ringing in a closed jar after most of the air is pumped out.**

Reason (R): Sound requires a medium to travel.

(i) Both A and R are true, but R is not the correct explanation of A.

(ii) Both A and R are true, and R is the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

✔ ANSWER

(ii) Both are true. We cannot hear the bell because there is almost no air (medium) left for the sound to travel through.

10.3 Sound Waves · Activity

Activity 10.5 **Push and pull one end of a slinky with a marked turn. Do you observe a disturbance moving towards your friend? Are a series of disturbances produced and do they move along the slinky? Does the mark move back and forth parallel to the direction of the disturbance?**

✔ ANSWER

- Yes, a disturbance (turns closer together) moves towards the friend.
- Repeated pushes and pulls make a series of disturbances: regions where turns are close and regions where they are spread out, moving along the slinky.
- The mark does not travel along the slinky. It only moves back and forth about its place, **parallel** to the direction of the disturbance.

Conclusion: The disturbance travels, the medium does not. Sound travels in a similar way (compressions and rarefactions).

Pause and Ponder · 10.3 Sound Waves

4 **Assertion (A): Compressions and rarefactions move through the medium.**
Reason (R): Individual particles of the medium continuously move forward with the

wave.

(i) Both A and R are true, but R is not the correct explanation of A.

(ii) Both A and R are true, and R is the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

✔ ANSWER

(iii) A is true, but R is false. The particles only vibrate about their mean positions; they do not travel with the wave.

10.4 Energy of Sound Waves · Activity

Activity 10.6 **Stretch a sheet over a container, sprinkle grains on it and make a loud sound nearby without touching it. Does the sound have any effect on the grains? Why does this happen?**

✔ ANSWER

- The grains move and jump. A louder sound makes them jump higher.
- Sound travels through air, reaches the sheet and makes it vibrate. The vibrating sheet moves the grains.

Conclusion: Sound is a form of energy; sound waves carry energy through the medium.

⭐ Pause and Ponder · 10.4 Energy of Sound Waves

5 **When sound travels from a tuning fork to your ear, which of the following actually reaches your ear?**

(i) Air particles near the tuning fork

(ii) Energy carried by sound waves

(iii) The tuning fork material

(iv) A continuous stream of compressed air

✔ ANSWER

(ii) Energy carried by sound waves. The air particles only vibrate about their places; it is the energy that is passed on.

⭐ Pause and Ponder · 10.5 Graphical Representation of a Sound Wave

6 **Fig. 10.17 (a) and (b) show the variation of density of the medium for two sound waves. Label compression and rarefaction by C and R. In (c) and (d), label the axes and draw the curves for (a) and (b).**

💡 SAMPLE ANSWER

- Mark **C** over every region where the dots are crowded (higher density) and **R** over every region where the dots are spread out (lower density).
- Label the x-axis **Distance** and the y-axis **Density**. Draw a dashed line for the average density.

- Draw a smooth wave: a **crest** above each C and a **trough** below each R.
- The wave with more crowded regions in the same length has a shorter wavelength, so its curve has more crests. (check the picture in your book)

10.6.1 Wavelength, frequency and time period · Activity

Activity 10.7 **Using a frequency app, sing or play 'Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa' and note each frequency. Compare each with 'Sa' by taking the ratio. Do you observe any pattern? Compare voice and mobile notes.**

💡 SAMPLE ANSWER

- The frequency is lowest for 'Sa' and increases for each next note.
- Ratios with 'Sa' are roughly: Re ≈ 1.12, Ga ≈ 1.25, Ma ≈ 1.33, Pa = 1.5, Dha ≈ 1.67, Ni ≈ 1.87, upper Sa = **2** (double the frequency).
- Voice and phone give nearly the same frequencies for the same note (voice values vary a little).

Each note has its own frequency.

Pause and Ponder · 10.6.1 Wavelength, frequency and time period

7 **Conduct Activity 10.1 with a thick rubber band and then with a thin rubber band. Does the thin rubber band vibrate faster? If yes, how do the frequency and time period of its sound differ from that of the thick rubber band?**

✔ ANSWER

Yes, the thin rubber band vibrates faster.

So its sound has a **higher frequency** and a **shorter time period** (v = 1/T) than that of the thick rubber band.

8 **If the frequency of a sound wave produced by an oscillating piston of a long tube filled with air is 20 Hz, how many oscillations does the piston complete per minute?**

✔ ANSWER

Given: v = 20 Hz = 20 oscillations per second; 1 minute = 60 s

Oscillations = 20 × 60

= 1200 oscillations per minute

9 **For the sound wave represented by the graph in Fig. 10.19 (density vs distance, marks at 0, 1.5, 3.0, 4.5 cm), what is half of its wavelength?**

✔ ANSWER

The curve crosses the average-density line at 1.5 cm, 3.0 cm and 4.5 cm. One full wave (crest + trough) takes 3.0 cm.

λ = 3.0 cm

Half of wavelength = **1.5 cm**

10.6.3 Speed of Sound · What if ...

What if ... **What if the speed of sound in air depended on its frequency? Would music still sound pleasant when a singer performs with instruments? Why or why not?**

💡 SAMPLE ANSWER

No. High and low notes would reach the listener at different times.

The voice and instruments (and even the different frequencies in one note) would get out of step, so the music would sound jumbled, especially far from the stage.

Pause and Ponder · 10.6.3 Speed of Sound

10 **Table 10.1: Speed of sound at 15 °C**

State	Medium	Speed
Solid	Steel	5000 m s^{-1}
Liquid	Water	1500 m s^{-1}
Gas	Air	340 m s^{-1}

Find the ratio of (i) speed in water to speed in air, (ii) speed in steel to speed in water.

✔ ANSWER

(i) 1500 ÷ 340 ≈ **4.4** (sound is about 4.4 times faster in water than in air)

(ii) 5000 ÷ 1500 ≈ **3.3** (sound is about 3.3 times faster in steel than in water)

11 **Two friends stand along a steel fence 340 m apart. Gunjan puts her ear on the fence and her friend knocks it. Using Table 10.1, find the time difference between the sound reaching her through air and through steel. Could she hear them separately? (Gap must be at least 0.1 s.)**

✔ ANSWER

Formula: time = distance ÷ speed

Through air: t_1 = 340 m ÷ 340 m s^{-1} = 1 s

Through steel: t_2 = 340 m ÷ 5000 m s^{-1} = 0.068 s

Difference = 1 − 0.068 = **0.932 s**

Since 0.932 s > 0.1 s, **yes**, she can hear the two sounds separately (first through steel, then through air).

10.6.4 Human perception of sound · Activity

Activity 10.8 **Play sounds from 100 Hz up to 1000 Hz in steps of 100 Hz with a mobile app. How does the sound change? Then reduce from 50 Hz to about 20 Hz.**

✔ ANSWER

- As the frequency increases, the sound becomes higher in pitch (shriller).
- As the frequency is reduced towards 20 Hz, the sound becomes deeper and fainter, and near 20 Hz it can hardly be heard.

Conclusion: Pitch depends on frequency, and humans hear only from about 20 Hz to 20 kHz.

Pause and Ponder · 10.7.1 Echo

12 **An experiment needs echoes to arrive at least 0.2 s after the sound is emitted. What minimum distance should a reflecting surface be placed at? Speed of sound = 343 m s^{-1}.**

✔ ANSWER

Sound goes to the surface and back, so distance = (v × t) ÷ 2

= (343 m s^{-1} × 0.2 s) ÷ 2 = 68.6 m ÷ 2

= 34.3 m

⭐ 10.8.1 Echolocation · What if ...

What if ... **What if humans could detect ultrasonic waves like dogs can? What would be the advantages and disadvantages?**

💡 **SAMPLE ANSWER**

Advantages: we could hear bats, dolphins and machines, maybe sense obstacles in the dark (like echolocation) and notice faults in machines early.

Disadvantages: many more sounds would disturb us (ultrasonic cleaners, sensors, animals), causing noise, stress and poor sleep.

⭐ Pause and Ponder · 10.8 Ultrasonic and Infrasonic Waves

13 **A sonar signal sent to find the depth of the ocean takes 4 s to return. What is the depth if the speed of sound in seawater is 1500 m s^{-1}?**

✔ **ANSWER**

Signal goes down and comes back, so depth = (v × t) ÷ 2

= (1500 m s^{-1} × 4 s) ÷ 2 = 6000 m ÷ 2

Depth = 3000 m (3 km)

⭐ Revise, Reflect, Refine

1 **Which observation best supports the idea that sound is a mechanical wave?**

(i) Sound shows reflection

(ii) Sound needs a medium to propagate

(iii) Sound has frequency

(iv) Sound carries energy

✔ ANSWER

(ii) Sound needs a medium to propagate. Waves that need a material medium are mechanical waves.

2 **For a sound wave propagating in a medium, increasing its frequency will increase its**

(i) wavelength

(ii) speed

(iii) number of compressions per second

(iv) time period

✔ ANSWER

(iii) number of compressions per second. Speed stays the same; wavelength and time period decrease.

3 **If 20 compressions pass a point in 4 seconds, the frequency is**

(i) 80 Hz

(ii) 5 Hz

(iii) 10 Hz

(iv) 0.2 Hz

✔ ANSWER

Frequency = 20 ÷ 4 s = 5 Hz

(ii) 5 Hz

4 **In a room, the reflected sound reaches the ear 0.05 s after its production. Will it produce an echo or reverberation? Justify your answer.**

✔ ANSWER

It will produce **reverberation**, not an echo.

An echo needs a gap of at least 0.1 s. Here the gap is only 0.05 s, so the reflected sound mixes with the original sound and makes it persist, as happens with quick multiple reflections in a room.

5 **Graphs of two sound waves are given in Fig. 10.30 (same scales). Which wave has (i) greater wavelength, and (ii) smaller amplitude? [(a): about 3 waves, shorter peaks; (b): about 6 waves, taller peaks]**

✔ ANSWER

(i) **Wave (a)** has the greater wavelength: it has fewer crests in the same distance.

(ii) **Wave (a)** has the smaller amplitude: its crests rise less above the average-density line than those of (b).

6 **The sound waves from three sources A, B and C are shown in Fig. 10.31. If the frequency of A is maximum and C is minimum, identify the curves and mark A, B and C.**

✔ ANSWER

Same speed, so higher frequency means shorter wavelength (more crests in the same distance).

- **A** = curve with the most crests (green curve)
- **B** = curve with the middle number of crests (red curve)
- **C** = curve with the fewest crests, longest wavelength (blue curve)

7 **Draw a graph to represent a sound wave for which the density amplitude is 3 units and wavelength is 4 cm.**

✔ ANSWER

- x-axis: Distance (cm); y-axis: Density. Draw a dashed line for the average density.
- Start on the average line at 0 cm.
- Crests 3 units above the line at 1 cm, 5 cm, 9 cm ...

- Troughs 3 units below the line at 3 cm, 7 cm, 11 cm ...
- The curve crosses the average line at 0, 2, 4, 6, 8 cm ...
- Two crests are 4 cm apart (= wavelength).

8 **In a movie, the explosion of a spacecraft in space shows a flash of light along with sound at the same time. What are the errors in this depiction?**

✔ ANSWER

- **Error 1:** Space is a near vacuum, so sound cannot travel there. No sound of the explosion should be heard at all.
- **Error 2:** Even where there is a medium, light travels much faster than sound, so the flash would be seen before the sound is heard, not at the same time.

9 **A source produces a sound wave of wavelength 3.44 m. If the wave travels with a speed of 344 m s^{-1}, find its time period.**

✔ ANSWER

Given: $\lambda = 3.44$ m, $v = 344$ m s^{-1}

$v = \lambda/T$, so $T = \lambda/v$

$T = 3.44$ m $\div$ 344 m s^{-1}

T = 0.01 s

10 **A ship sent a sonar signal and detected an echo after 5 s. If the ultrasonic wave travels at 1525 m s^{-1} in seawater, how far down is the wreckage of the sunken ship?**

✔ ANSWER

Given: t = 5 s (down and back), v = 1525 m s^{-1}

Depth = (v × t) ÷ 2 = (1525 × 5) ÷ 2 = 7625 ÷ 2

Depth ≈ 3812.5 m (about 3.8 km)

11 **A vehicle's ultrasonic parking sensor (about 40 kHz) starts beeping at 1.2 m from an obstacle. How much time does the ultrasonic wave take to go to the obstacle and come back? Speed = 345 m s^{-1}.**

✔ ANSWER

Total distance = 2 × 1.2 m = 2.4 m

time = distance ÷ speed = 2.4 m ÷ 345 m s^{-1}

t ≈ 0.007 s (about 7 milliseconds)

12 **The speed of sound in air is about 331 m s^{-1} at 0 °C and nearly 344 m s^{-1} at 22 °C. Roughly how much extra time will thunder take to travel 1720 m if the temperature changes from 22 °C to 0 °C?**

✔ ANSWER

At 22 °C: t_1 = 1720 m ÷ 344 m s^{-1} = 5 s

At 0 °C: t_2 = 1720 m ÷ 331 m s^{-1} ≈ 5.2 s

Extra time = 5.2 − 5 = **about 0.2 s**

13 **Fig. 10.32 shows the variation of density for a sound wave travelling at 340 m s^{-1} (an 8 cm length is marked on it). Calculate the wavelength and frequency.**

✔ ANSWER

The 8 cm runs from the middle of the 1st compression to the middle of the 3rd compression, so it covers two full wavelengths.

λ = 8 cm ÷ 2 = 4 cm = 0.04 m

v = v ÷ λ = 340 m s^{-1} ÷ 0.04 m

λ = 4 cm, v = 8500 Hz

14 **Fig. 10.33 shows two sound waves A and B travelling at 345 m s^{-1} (distance marks at 0, 2.5, 5.0 cm). Find the wavelength of each and their frequencies.**

✔ ANSWER

From the graph: A repeats every 2.5 cm; B repeats every 5.0 cm.

Wave A: λ = 2.5 cm = 0.025 m; v = 345 ÷ 0.025 = **13800 Hz**

Wave B: λ = 5.0 cm = 0.05 m; v = 345 ÷ 0.05 = **6900 Hz**

15 **Two identical sound sources at A (in air) and B (in water) produce sound at the same time;**

it travels horizontally to a vertical cliff and comes back. If the time to return to A is 4.5 times that of B, what is the ratio between the speeds of sound in air and water?

✔ ANSWER

Both travel the same distance 2d (to the cliff and back).

v = 2d/t, so speed is inversely proportional to time.

$v_{air} \div v_{water} = t_B \div t_A = 1 \div 4.5$

$v_{air} : v_{water} = 1 : 4.5 = 2 : 9$

⭐ The Journey Beyond

1 **Find out about research on the effect of excessive earphone use on hearing, how hearing is tested, the decibel ranges for mild, moderate and severe hearing loss, and government schemes for hearing aids and cochlear implants. Write an article.**

💡 SAMPLE ANSWER

Points for your article:

- Listening loud for long hours through earphones can damage the inner ear and cause hearing loss and ringing in the ears. Keep volume low and take breaks.
- Hearing is tested with an audiogram (by an audiologist).

- Find the dB ranges for mild, moderate and severe hearing loss from a reliable health website.
- Look up the Government of India ADIP scheme for aids and appliances, including cochlear implants for young children.

2 **Cover a mobile phone playing music with a paper cone. Compare the loudness with and without the cone. Try different shapes.**

💡 **SAMPLE ANSWER**

The sound is louder in the direction the cone points, because the cone guides the sound energy in one direction instead of letting it spread in all directions. A longer cone with a wide mouth usually works better.

3 **How does the curved design of ceilings and walls behind the stage in concert halls improve sound for the audience compared to flat surfaces?**

💡 **SAMPLE ANSWER**

Curved surfaces reflect sound from the stage towards the audience and spread it evenly over the hall, so people at the back also hear clearly. Flat surfaces may send sound in unwanted directions or cause echoes.

4 **Measure the speed of sound in a large ground by bursting a balloon and timing the gap between seeing and hearing it. What value did you get? Compare with about 346 m s^{-1} at 25 °C. (vi) Why did you measure the time between 'seeing' and 'hearing' the balloon burst?**

💡 **SAMPLE ANSWER**

Speed = distance ÷ average time. For example, 200 m ÷ 0.6 s ≈ 333 m s^{-1}, close to 346 m s^{-1} (errors come from reaction time).

(vi) Light reaches us almost instantly, so seeing the burst marks the moment the sound starts. The gap till we hear the 'pop' is the time sound takes to travel the distance.

5 **Explore internet resources (PhET Sound Waves, Chrome Music Lab, phyphox) on the effect of humidity and temperature on the speed of sound.**

💡 **SAMPLE ANSWER**

You will find that the speed of sound in air increases when temperature or humidity increases (e.g. about 331 m s^{-1} at 0 °C and about 344 m s^{-1} at 22 °C).

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**